

Reflective Journal

Completing this lab helped me better understand how the different tools used in machine learning work together. Before working through the exercises, programs such as Pandas, NumPy, Matplotlib, and Jupyter Notebook seemed like separate tools that I would have to learn individually. As I worked with the Iris dataset, I started to understand that each tool has a specific purpose within the same overall process. Jupyter Notebook provides the environment to write and test the code, Pandas organizes the data, NumPy performs calculations, and Matplotlib helps turn the data into something that can be visually interpreted. Seeing these tools interact made the machine learning workflow seem less intimidating because I could understand the role each part played.

One of the biggest things I learned was how useful Pandas can be for working with structured data. The DataFrame reminded me of a spreadsheet because the information was organized into rows and columns, but I began to see that Pandas provides much more control over the data. For example, I could select a specific column, separate records based on species, calculate averages, or count values without manually going through each row. At first, commands such as `df['species']` or filtering the DataFrame based on a condition did not feel as natural as selecting cells in Excel. However, after seeing the results, the logic became clearer. Instead of manually finding information, I could tell Python exactly what group or column I wanted to examine. This helped me understand why Pandas is valuable when datasets become much larger than something that could be reviewed by hand.

My understanding of NumPy also improved when I used it to calculate the mean and standard deviation of the sepal lengths. I had previously thought of NumPy mostly as another Python library that had to be imported, but this exercise showed me how it connects programming with mathematical analysis. Instead of calculating each individual value, NumPy could work with an entire collection of measurements at once. This made the idea of performing mathematical operations on data feel much more practical. I can see how this becomes important in machine learning because models may need to process thousands or even millions of values. Being able to perform calculations across an entire set of data is much more efficient than trying to handle each value separately.

I also became more comfortable with the idea of using an interactive environment such as Jupyter Notebook. Being able to run one section of code and immediately see the result made it easier to understand what each command was doing. If something did not produce the result I expected, I could focus on that individual section instead of trying to troubleshoot an entire program at once. This approach made Python feel more manageable and encouraged me to experiment with the code rather than being afraid of making a mistake.

Overall, the most important thing I learned from this lab was that machine learning is built from several smaller skills that work together. Organizing data, performing calculations, visualizing relationships, and documenting the results are all necessary before moving into more advanced machine learning techniques. I now have a better understanding of why these tools are commonly used together and how they can help turn raw information into something that can be analyzed. As I continue into more advanced labs, I think this foundation will make it easier to understand not only how to run machine learning code, but also why each step in the process matters.